

Discussion section 2

Huangchen Zhou

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1 Chp 2, Q15

Assume that $X_0, X_1, \dots$ is a two-state Markov chain on $\mathcal{S} = \{0, 1\}$ with transition matrix

$$P = \begin{matrix} & \begin{matrix} 0 & 1 \end{matrix} \\ \begin{matrix} 0 \\ 1 \end{matrix} & \begin{pmatrix} 1-p & p \\ q & 1-q \end{pmatrix} \end{matrix}.$$

The present state of the chain only depends on the previous state. One can model a bivariate process that looks back two time periods by the following construction. Let $Z_n = (X_{n-1}, X_n)$, for $n \geq 1$. The sequence $Z_1, Z_2, \dots$ is a Markov chain with state space $\mathcal{S} \times \mathcal{S} = \{(0, 0), (0, 1), (1, 0), (1, 1)\}$. Give the transition matrix of the new chain.

Proof. We are going to compute the transition matrix, say Q of the new chain, where each entry, for each $i, j, k, l \in \mathcal{S}$ is given by

$$\begin{aligned} Q_{(i,j),(k,l)} &= \mathbb{P}(Z_2 = (k, l) | Z_1 = (i, j)) \\ &= \mathbb{P}(X_1 = k, X_2 = l | X_0 = i, X_1 = j). \end{aligned}$$

If $j \neq k$, then $Q_{(i,j),(k,l)} = 0$.

If $j = k$, then

$$\begin{aligned} Q_{(i,j),(k,l)} &= \mathbb{P}(X_1 = j, X_2 = l | X_0 = i, X_1 = j) \\ &= \mathbb{P}(X_2 = l | X_0 = i, X_1 = j) \\ &= \mathbb{P}(X_2 = l | X_1 = j) = P_{jl} \end{aligned} \tag{1.1}$$

where (1.1) uses the memoryless property of the Markov chain $\{X_n\}_{n \geq 0}$.

To summarize, the transition matrix Q of the new chain is given by

$$Q = \begin{matrix} & \begin{matrix} (0,0) & (0,1) & (1,0) & (1,1) \end{matrix} \\ \begin{matrix} (0,0) \\ (0,1) \\ (1,0) \\ (1,1) \end{matrix} & \begin{pmatrix} P_{00} & P_{01} & 0 & 0 \\ 0 & 0 & P_{10} & P_{11} \\ P_{00} & P_{01} & 0 & 0 \\ 0 & 0 & P_{10} & P_{11} \end{pmatrix} \end{matrix} = \begin{matrix} & \begin{matrix} (0,0) & (0,1) & (1,0) & (1,1) \end{matrix} \\ \begin{matrix} (0,0) \\ (0,1) \\ (1,0) \\ (1,1) \end{matrix} & \begin{pmatrix} 1-p & p & 0 & 0 \\ 0 & 0 & q & 1-q \\ 1-p & p & 0 & 0 \\ 0 & 0 & q & 1-q \end{pmatrix} \end{matrix}.$$

□

2 Chp 3, Q7

A Markov chain has n states. If the chain is at state k , a coin is flipped, whose heads probability is p . If the coin lands heads, the chain stays at k . If the coin lands tails, the chain moves to a different state uniformly at random. Exhibit the transition matrix and find the stationary distribution.

Proof. Denote the state space by $\mathcal{S} = \{1, 2, \dots, n\}$. For each $l \in \mathcal{S}$,

$$\mathbb{P}(X_1 = l | X_0 = k) = \begin{cases} p & \text{if } l = k \\ (1-p) \cdot \frac{1}{n-1} & \text{if } l \neq k. \end{cases}$$

In other words, the transition is a matrix with value p on the diagonal, and is equal to $(1-p) \cdot \frac{1}{n-1}$ off diagonal.

To find the stationary distribution π , it suffices to solve for $\pi = (\pi_1, \pi_2, \dots, \pi_n)$ such that $\pi P = \pi$, with $\pi_1 + \pi_2 + \dots + \pi_n = 1$.

By looking at the first component of the product $\pi P = \pi$, we have

$$\pi_1 \cdot p + (\pi_2 + \pi_3 + \dots + \pi_n) \cdot (1-p) \cdot \frac{1}{n-1} = \pi_1.$$

Plugging $\pi_1 + \pi_2 + \dots + \pi_n = 1$ into the above equation we have

$$\pi_1 \cdot p + (1 - \pi_1) \cdot (1-p) \cdot \frac{1}{n-1} = \pi_1.$$

Thus $\pi_1 = 1/n$. Similarly $\pi_i = 1/n$ for each $i = 1, \dots, n$. Thus the stationary distribution is $\pi = (1/n, \dots, 1/n)$. □